

Spatiotemporal Modelling of Coastal Vulnerability and Ecosystem Degradation in Indus Delta: A Landsat and DSAS Approach

Dr. Muhammad Irfan ^{a*}, Shahzaib Farrukh^a, Dr. Humaira Altaf Khan^b, Dr. Shaista Mahmud^a, Bassam Nasir^c and Mohammad Idrees^d

^a Department of Earth & Environmental Sciences, Bahria University, Pakistan.

^b Department of Mathematics, NED University of Engineering and Technology, Pakistan.

^c Department of Civil, Construction and Environmental Engineering University of Central Florida.

^d Department of Maritime Sciences, Bahria University, Pakistan

*Corresponding author: Muhammad Irfan, Email: mirfan.bukc@bahria.edu.pk

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ABSTRACT

The Indus Delta in Pakistan, one of the largest arid climate deltas, is experiencing severe ecological and geomorphological degradation due to climate change and unsustainable human activities. This study presents a multi-temporal assessment of critical environmental issues, including shoreline retreat, sediment flux reduction, sea-level rise and vegetation loss, by utilizing Geographic Information Systems (GIS), remote sensing and predictive modelling by integrating DSAS-based shoreline dynamics with Normalized Difference Vegetation Index (NDVI) and Suspended Sediment Index (NDSSI).

Using Landsat 8 and 9 satellite datasets from 2013 to 2023, the study reveals alarming trends: the shoreline is retreating at an average rate of -24.1 m/year, with a net shoreline movement of -226.4 meters, indicating significant coastal erosion. Vegetation cover has plummeted by 55%, from 4,200.21 km² in 2013 to 1,879.20 km² in 2023, including a 20% loss in mangrove areas and a 33% decline in species diversity, threatening the delta's ecological integrity. Additionally, concentration has decreased by 25%, largely due to upstream dam construction and river diversions.

Sea level projections indicate a rise of 25 cm by 2050, and vulnerability modeling predicts that a 10-meter storm surge could inundate 9,376 km² (71%) of the delta by 2070. These findings highlight a threshold where ecosystem degradation and biodiversity loss undermine coastal resilience and socioeconomic sustainability.

The study emphasizes the urgent need for integrated coastal zone management to rehabilitate mangrove ecosystems, restore sediment flows, and ensure sustainable freshwater governance. It provides a geospatial framework crucial for climate adaptation planning and policy development in one of South Asia's most threatened delta systems.

1. Introduction

Socioeconomically, the delta is vital for millions of people who depend on its fertile land and water resources. However, its sustainability is under severe threat from coastal erosion, mangrove extinction, and the marine ecosystem degradation. Deltas are critical

interfaces where continental and marine ecosystems interact to support the biodiversity of and marine ecosystem. The Indus Delta, located in southern Pakistan, is the sixth-largest in the world, and covers approximately 41,440 km², ranked as the sixth-largest deltaic zone [1]. The delta hosts diverse fauna and flora such as dolphins, migratory birds, and mangrove

forests. The region is also used for commercial agriculture and fishing activities [2]. The deltas, a distinct coastal feature around the world, characterize a critical interface between continental and marine ecosystems. It provides significant economic, ecological, and social benefits [3]. However, deltaic regions are highly vulnerable to climate change impacts, mainly rising sea level, and anthropogenic stresses such as land reclamation, dam structure, and unsustainable resource exploitation [4].

The Indus Delta, Pakistan, is one of the world's vulnerable arid climate deltas, experiencing rapid environmental degradation due to natural and human-induced factors [5]. The sediment supply plays a key role in sustaining the deltaic integrity and ecological health. The upstream development activities (dam construction, diversion of canals for irrigation) significantly reduced the sediment flux, aggravating habitat loss and coastal erosion [3]. The literature shows that a 90% decline in sediment inflow to the delta compared to past river flow greatly affects nutrient distribution in the coastal zone for fish productivity and mangrove health [6,7]. Satellite data analysis and GIS have established an important tool for monitoring temporal and spatial environmental degradation in the deltaic zone. The concentration of sediment and vegetation health are mainly evaluated by employing NDVI and NDSSI, respectively, which reveal a vital insight for ecosystem dynamics [8]. These tools have been used in diverse coastal zones, including the Indus delta, illustrating a decline in mangrove due to sediment starvation and seawater intrusion [9].

Globally, shoreline fluctuations and erosion rates have been computed effectively by using the Digital Shoreline Analysis System (DSAS) [10,11]. DSAS application in the Indus Delta represents an alarming drift of shoreline retreat and approximately 300 km² land loss over 50 years [12]. The land loss directly threatens the biodiversity, and the socioeconomics of local communities depend on coastal resources. Rising sea level is mainly driven by glacier melt and global warming. This rise has contributed to the loss of agricultural lands, seawater intrusion into freshwater zones, mangrove extinction, and a high probability of storm surges [7]. Integrated multi-temporal geospatial studies combining satellite imagery, shoreline analysis, and predictive modeling are critical to developing sustainable coastal management strategies. Such methods help to classify vulnerable areas, compute degradation patterns, and recommend restoration interventions [8].

Sea level rise and land degradation are two of the primary issues faced by the coastal communities [13].

The important factors leading to land degradation include coastal erosion, reduced organic content, soil compaction, salinization, and loss of biodiversity [14]. Among these factors, coastal erosion stands out as a global phenomenon, and researchers and investigators are explicitly concerned about the rising issue of coastal erosion [4,9,15]. The erosion process is dependent on natural and anthropogenic activities occurring in that area [16]. Coastal erosion occurs over the long-term sediment subtraction by the action of waves, currents, and human interventions [17]. Similarly, natural processes causing coastal erosion in the Indus Delta are tidal action, sea level rise, and seawater intrusion [7]. These three are significant contributing factors to the increasing vulnerability of coastal communities to natural hazards, and destroying the natural ecology by degrading the mangroves [18]. Mangrove forest acts as a natural barrier to erosion, yet over the past 100 years, mangrove cover in Pakistan has decreased due to rising sea levels and anthropogenic activities, whereas in the last two decades, restoration of mangroves has been reported by numerous studies [19,20]. The sea level is rising approximately 3.7 mm/year according to regional trends and global observation [21]. The predictive modelling forecast a potential rise in sea level, i.e., 25 cm by 2050. The storm surges inundate a large region of the deltaic zone by 2070, threatening both human settlement and the ecosystem.

Human activities have altered the discharge pattern and directly influence sediment transport, resulting in the deltas receiving low influx [22]. Four general types have been classified for engineering development that have been adopted along the Indus River. (i) Channels to provide the water for irrigation; (ii) Barrages to divert river flow into channels; (iii) Embankments & dikes stopping overflow and thereby controlling flow to the main canals; and (iv) Dams for hydro-electric energy and flood control. The wide-scale modification of the Indus initiated in the 1940s with the construction of barrages and diversion channels. Consequently, the river discharge and sediment load downstream decreased considerably [23]. The river discharge in the deltaic region has now been reduced to about one-fifth, and the river has been confined to a single channel down to the coastal area. The present active delta has shrunk to a small triangular area [5]. The freshwater discharge in Kotri Barrage has reduced from 146 million acre feet per year (MAF/year) to less than 10 MAF/year, causing up to 90% sediment flux reductions compared to historical levels [24]. These changes have initiated extensive shoreline retreat,

salinization, mangrove loss, and reduced fish stocks [3].

High-resolution satellite data integration with GIS has been extensively utilized to monitor the shoreline fluctuations in deltaic regions [25, 26]. The innovation in satellite technology offers an effective tool for capturing spatial and temporal changes across the globe in the dynamic coastal system. The effective monitoring has already been covered successfully for shoreline movements in Turkey's Ramsar wetlands [27,28], India's Kannyakumari coastline [29], and Egypt's North Sinai coast [30]. The geomorphological changes are detected efficiently by using the remote sensing imagery for Bangladesh [31] and the Mediterranean shoreline [32]. The reliability of data is crucial to provide frequent updates and wide area analysis from satellite imagery in order to begin smart monitoring for short and long-term fluctuations on the shoreline. The shoreline variations have been assessed by applying multiple tools in remote sensing, such as the edge detection approach [33], spectral band ratios [34], and image classification [35]. Remote sensing methods provide effective results for the analysis of China's Yellow River Delta and shorelines [36,37]. Previous studies also applied the Google Earth Engine, which successfully integrated the machine learning approaches to map the random forest and quantify the land cover precisely, shoreline retreat investigation, and mangrove mapping to estimate the density and aerial extent [38,39]. This study utilized multi-temporal satellite imagery (Landsat 8 and 9) from 2013-2022, coupled with indices, the NDVI, NDSSI, and DSAS, to evaluate sediment concentration decline, shoreline retreat, and vegetation loss. The analysis provides an understanding of the interplay between climate drivers and human interventions contributing to delta degradation. The outcome of research focused on addressing geospatially informed regions for climate adaptation and ecosystem restoration.

2. Study Area

The Indus Delta is situated in southern Pakistan, a distinct geomorphological feature shaped in an arid climate. It ranks as one of the world's biggest deltas, identified as the fifth largest by some authors [40,41], and ranked as sixth by others [3,42]. The River Indus drains into the north-eastern Arabian Sea, developing a large fan-shaped delta. The freshwater and nutrient-rich sediment load has a great influence on marine species of the Indus Estuary and adjacent coastal regions [43]. The medium and fine sediments of the

Indus River build up the delta, which has prograded seaward approximately 80 km in 2,000 years.

The deltaic coastal morphology comprised a network of tidal creeks formed due to a shift in channels and a large number of small islands with sparse mangrove vegetation [44]. The Indus canyon runs north-east to south-west, partially intersecting the continental shelf about 30 km from the Indus Deltaic shore [45]. Coastal dynamics such as strong tidal currents, wind, and wave action create coastal erosion; variations in the suspended material and river discharges; littoral transport and tectonic activities – subsidence and uplift of the delta [46]. A network of creeks extends over 170 km in the south-east of Karachi. These creeks meander about 50 to 60 km inland. Seawater flows in the creeks with velocities as high as 2.3 m/s during the flood and ebb tides. The major creeks of the Indus Delta are marked in Figure 1.

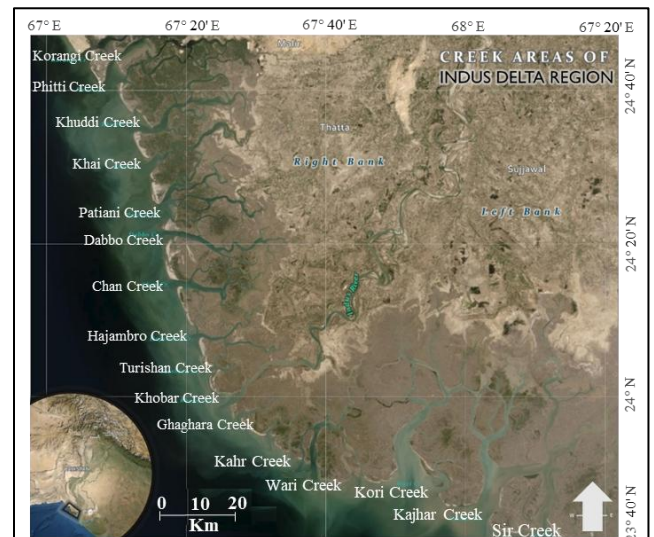


Fig. 1. Location map of the Indus Delta, Pakistan.

The region of the deltaic plain from the shoreline to the alluvial valley covers about 29500 km² in a fan shape. It characterizes a typical dry (arid) subtropical delta with negligible precipitation and a high evaporation rate. The arid climate, together with a moderate to high tide range along the coast, has produced halite-encrusted mud flats. In the lower deltaic plain, substantial storm tides from southwest monsoon winds in summer inundate with salt water vast areas of both the active and the abandoned deltaic plains. The total area of mud flats in the Indus Delta is estimated to be approximately 385,000 hectares. Out of this, about 200,000 hectares are covered with mangrove vegetation.

3. Methodology

The study utilized a multi-parametric geospatial approach coupled with remote sensing, shoreline trend

analysis, vegetation and sediment index derivation and inundation modelling to evaluate ecological degradation and compute shoreline retreat in the Indus Delta over the period from 2013-2023. The workflow comprises six key steps: data acquisition, vegetation analysis (NDVI), sediment load analysis (NDSSI), shoreline change analysis (DSAS), sea-level trend modeling by altimetry data, and vulnerability estimation using DEM-based simulations.

3.1 Data Acquisition and Preprocessing

Landsat 8 OLI for the period of 2013-2018 and Landsat 9 OLI-2 for 2023 were acquired from USGS with minimum cloud ($\leq 5\%$) and level 2 surface reflectance. These images contain WRS path 152 and Row 43 corresponding to the Indus Delta (Table 1). The dataset was initially processed for atmospheric and radiometric correction. The area of interest was extracted using the Mask tool in ArcGIS. The land cover, shoreline features, and geometric accuracy were ensured by cross-checking with time series high-resolution data from Google Earth.

Table 1

Landsat data acquisition and parameters

S#	Satellite	Date of Acquisition	Path	Row	Processing	Cloud
1	Landsat 8	2013-10-14	152	43	L2SP	1.00
2	Landsat 8	2018-10-12	152	43	L2SP	0.04
3	Landsat 9	2023-10-18	152	43	L2SP	4.80

3.2 Dynamics of Vegetation Cover using NDVI.

To evaluate the vegetation health and growth over time, the NDVI was quantified from surface reflectance values of red and near infrared (NIR) bands of the Landsat sensor, as in equation 1

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

Where Red and NIR characterize the reflectance values in red and near infrared bands (Bands 4 and 5) for Landsat 8 & 9. NDVI was estimated for the years 2013, 2018, and 2023, categorized into intervals for vegetation density, and the temporal trend for vegetation degradation.

3.3 Suspended Sediment Estimation via NDSSI

The concentration of suspended sediment was calculated using the NDSSI provides an effective index for optical turbidity quantification in the deltaic zone as expressed in equation 2.

$$NDSSI = \frac{\rho_{Red} - \rho_{Green}}{\rho_{Red} + \rho_{Green}} \quad (2)$$

Where Red defines Band 4 (0.64-0.67 μm) and Green defines Band 3 (0.53-0.59 μm). The values for NDSSI are positively associated with suspended sediment in water. The higher values show an increase in sediment load depicted by the red spectral region. The NDSSI map used to assess variation in sediment influx correlated with upstream hydrological barriers and climate-induced flow changes.

$$\Delta NDSSI = NDSSI_{2023} - NDSSI_{2018} \quad (2a)$$

A negative $\Delta NDSSI$ indicates sediment depletion using equation 2a.

3.4 Shoreline Change Analysis Using DSAS

Temporal shore shift was estimated using the Digital Shoreline Analysis System (DSAS) tool, an extension of ArcGIS designed by USGS. Digitization of shoreline for the years 2013, 2018, and 2023 imagery. A baseline was established ~ 2 km inland, and 1373 transects were created at 200 m intervals, extended perpendicular to the shoreline. The statistical metrics applied to calculate the Net Shoreline Movement (NSM), a linear distance between two shoreline points, equation 3. End Point Rate (EPR), annual shoreline changes over the period as defined in equation 4, and Linear Regression Rate (LRR), a best-fit linear model that measures long-term trends across shoreline positions as in equation 5. The generated results from DSAS allow the estimation of the spatially explicit rate of erosion/accretion with their distribution at the delta front.

$$NSM = S_{end} - S_{start} \quad (3)$$

$$ERP = \frac{NSM}{T} \quad (4)$$

Where T is the temporal interval in years.

$$y = \beta_0 + \beta_1 x + \epsilon \quad (5)$$

Where x is time, y is the shoreline position, β_1 is the slope (rate of change), and ϵ is residual error.

3.5 Rising Sea Level and Projection Modelling

Sea level dynamics were examined using the global altimetry dataset from Copernicus Climate Change Service (C3S) covering the years 1993–2020. The Net CDF data was imported into ArcGIS to run linear regression for mapping by using the tool (Generate Trend Raster). Moreover, the future sea level rise up to 2050 projection was computed and visualized from equation 6.

$$SLR_{future} = SLR_{observed} + (r \times \Delta t) \quad (6)$$

Where SLR observed is the sea level in the base year, e.g., 2020, r is the mean annual rate of sea level

rise in cm/year, and Δt is the number of years into the future as projected. This approach facilitated the modelling for the generation of projected regional sea level rise and well-versed effective subsequent inundation modeling.

3.6 Coastal Vulnerability and Inundation Risk Mapping

A coastal vulnerability map was created using a Digital Elevation Model (DEM) 30-meter resolution from the Shuttle Radar Topography Mission (SRTM). A total of four tiles were combined to encompass the entire Indus Delta. The DEM model was segmented into 10 elevation bands (1-10 m). To calculate the prospective inundated land due to a projected 10 m storm surge or rise in sea level, the equation was employed. Overlay analysis was carried out to identify the most susceptible zone within the Indus Delta and flooding under given climate conditions to provide spatial risk estimation and adaptive planning for future.

$$A_{inundated} = \sum_{i=1}^n A_i \text{ where } Z_i \leq H_{surge} \quad (7)$$

Where A_i is pixel area at elevation, and H_{surge} is modelled or projected storm surge height, i.e., 10 m, and Z_i is the value of DEM elevation for each cell.

The principal limitation of this research lies in the dataset, where the 30-meter spatial resolution of the Landsat 8 and 9 restricts the detection of micro-scale features. The overpass time of the satellite also limits the tidal variability for shoreline position uncertainty. Despite the limitations, the study evaluates the multi-decadal and multi-indicator by applying an effective approach that enables robust macro-scale assessments.

3.7 Accuracy Assessment & Validation

The accuracy assessment was carried out in multiple steps to ensure the reliability of results and analysis. The extraction of land cover and shoreline features from Landsat data was validated by utilizing high-resolution time series imagery data from Google Earth. The reference points were identified for visual comparison, mainly focusing on mangrove patches, water bodies, bare soil, and shoreline extents. The results at this step were evaluated using confusion matrices that rely on stratified random sampling and show accepted threshold values between reference data and classification (Table 2).

The NDVI was validated by using vegetation extent from classified maps with reference to mangrove coverage. The comparison shows a

deviation of around $\pm 5\%$ in dense vegetation, enhancing the reliability of NDVI results. NDSSI results were validated by using sediment concentration and turbidity observation of the study area. The strong to moderate correlation between high sediment load and high NDSSI represents the accuracy and consistency of the analysis (Table 2)

The shoreline variation generated by the DSAS tool was verified in two ways: (1) the shoreline extracted from Landsat images (2013, 2018, and 2023) was compared with the manually digitized shoreline from Google Earth imagery, (2) the results include a confidence interval (CI) to estimate uncertainty with the EPR with standard error ± 2.12 m/year, and the Net Shoreline Movement (NSM) provides a confidence interval of ± 16.87 m. These limits represent a statistical validation and consistency in shoreline change detection. Finally, sea level rise (SLR) was validated against the IPCC assessment report and projected model aligned with the global trend (Table 2). The results estimated from the above methodology show high accuracy and reliability of spatio-temporal coastal degradation in the Indus delta.

Table 2

Validation metric for results accuracy

Method	Validation Metric	Value
Land Cover	Overall Accuracy OA>85%	88.3%
Kappa Coefficient	Kappa Coefficient K>0.75	0.81
NDVI Validation	RMSE (compared to reference maps)	$\pm 5.4\%$
NDSSI Validation	R ² (vs. sediment hotspots)	0.74
Shoreline Analysis	EPR Std. Error	± 2.12 m/year
	NSM Std. Error	± 16.87 m
Sea-Level Model	Trend Agreement (vs. IPCC +3.7–4.2 mm/year)	+3.9 mm/year

4. Results & Discussion

The Indus delta is depleting with time due to the reduction of sediments supplied by the river discharge or other contributing factors. Hence, the Indus Delta is transforming into a wave-dominated delta. In this study, a large-scale quantitative assessment is conducted to calculate the land loss over the years in the creeks of the Indus Delta region.

4.1 Sea-Level Analysis

The “Predictive analysis” method of ArcGIS Pro is utilized to predict a steady acceleration in Sea Level Rise (SLR) across the Indus Delta (Figure 2). Modelling against the 1993–2023 satellite altimetry baseline, where the mean SLR is projected to rise approximately 25 cm by 2050. The forecast values are aligned with the latest Intergovernmental Panel on Climate Change (IPCC) sixth assessment report, i.e., 21–71 cm by 2070. The average 44 cm rise has been proposed by IPCC. The deltaic regions are the areas where the river opens into the sea, and these regions are rapidly sinking below the sea level. The probable cause is due to a lack of silt, decreased river flows, and human-induced activities. Therefore, these regions are becoming highly susceptible to climate change impacts.

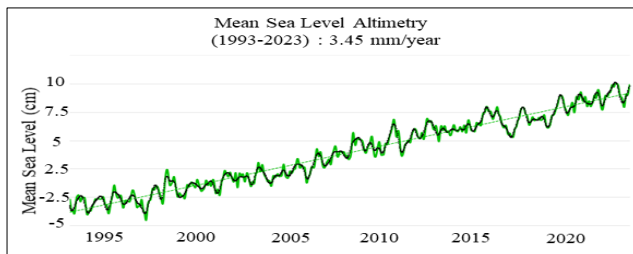


Fig. 2. Global Sea level rise graph from 1995-2020.

Table 3

Projected SLR for the Indus Delta

Year	Scenario / Source	Relative rise (cm)
2050	ArcGIS forecast (this current study)	+25
2070	IPCC sixth report assessment	+21 – 71 (mean +44)

A scenario-based risk map of the Indus Delta depicts projected flood extent regions that are at risk of coastal flooding. The map is showing scenarios from (1-10 m) high waves, Figure 3. Modeling indicates that a 10-meter tsunami or cyclone could result in extensive flooding across the Indus Delta. The delta's vulnerable region threatens 71% of the entire Indus Delta, which is approximately 9376 km² (Table 4). This event poses a severe threat to coastal populations, potentially exposing hundreds of livelihoods and lives across the affected regions.

Table 4

Sea level and inundation scenarios

Scenario	SLR/cyclone surge height (m)	Inundated area (km ²)	% of delta (41,440 km ²)
Present SLR trend to 2050	+0.25	1 030	2.5 %

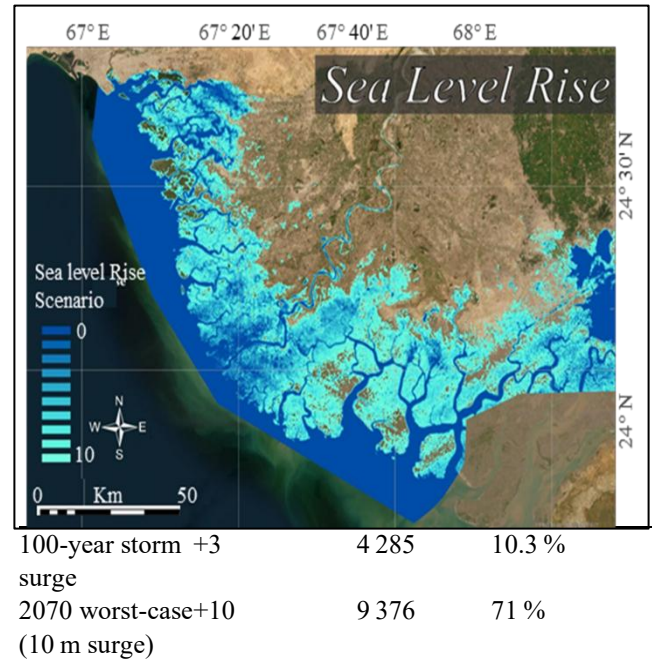


Fig 3. A scenario-based map of the Indus Delta showing regions that are at risk of coastal flooding.

4.2 The Concentration of Suspended Solids in the Indus Delta

To understand the Suspended Solid Concentration (SSC) in the Indus Delta, we applied the Normalized Difference Suspended Sediment Index (NDSSI) technique on satellite imagery for the years 2013, 2018, and 2023 (Figure 4). NDSSI is an effective tool for determining the concentrations of suspended sediment in the deltaic region, which is crucial in understanding sediment dynamics and coastal processes. The results of the NDSSI map (2013) indicate regions with lower concentrations of suspended sediments range from -0.931 to -0.219 along the coast and in the creeks, particularly in the central and eastern parts of the delta (Figure 4a). NDSSI map (2018) reveals a decline as compared to the NDSSI map (2013), and the highest value, 0.171, is estimated (Figure 4b). The decrease in SSC is due to a change in river discharge or sediment transport related to human activities. A minor recovery is observed in the 2023 map; however, the values are still negligible compared to the 2013 baseline.

The NDSSI map (2023) shows an increase in the maximum value of 0.159, indicating a confined rise in SSC compared to the 2018 higher values (Figure 4c). However, areas with low NDSSI values continue to expand, especially in the western deltaic region, suggesting that sediment concentrations remain insufficient due to anthropogenic activities such as upstream diversion and dam construction that restrict the natural processes. The spatio-temporal changes in NDSSI from 2013 to 2023 highlighted significant fluctuations in SSC throughout the Indus Delta (Table

5). These outcomes underline the critical need for consistent monitoring and the adoption of integrated sediment management tactics to alleviate sediment deficits, develop coastal resilience, and safeguard the sustainable development of the deltaic setting.

Table 5

NDSSI statistics for the Indus Delta

Year	NDSSI min	NDSSI max	Interpretation
2013	-0.931	0.219	Higher SSC in central/eastern creeks
2018	-0.891	0.171	Basin-wide value drop in SSC
2023	-0.882	0.159	Slight retrieval, still below the 2013 reference value

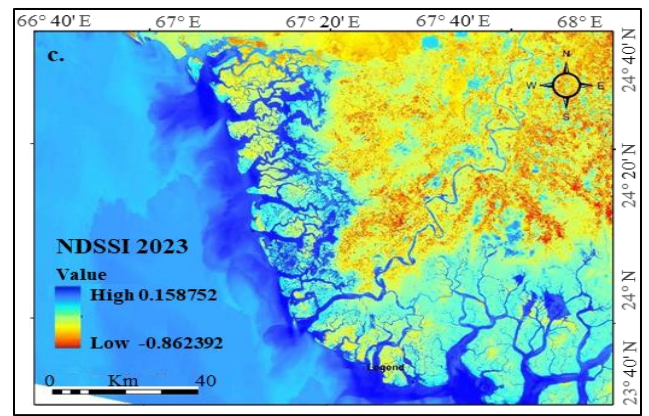
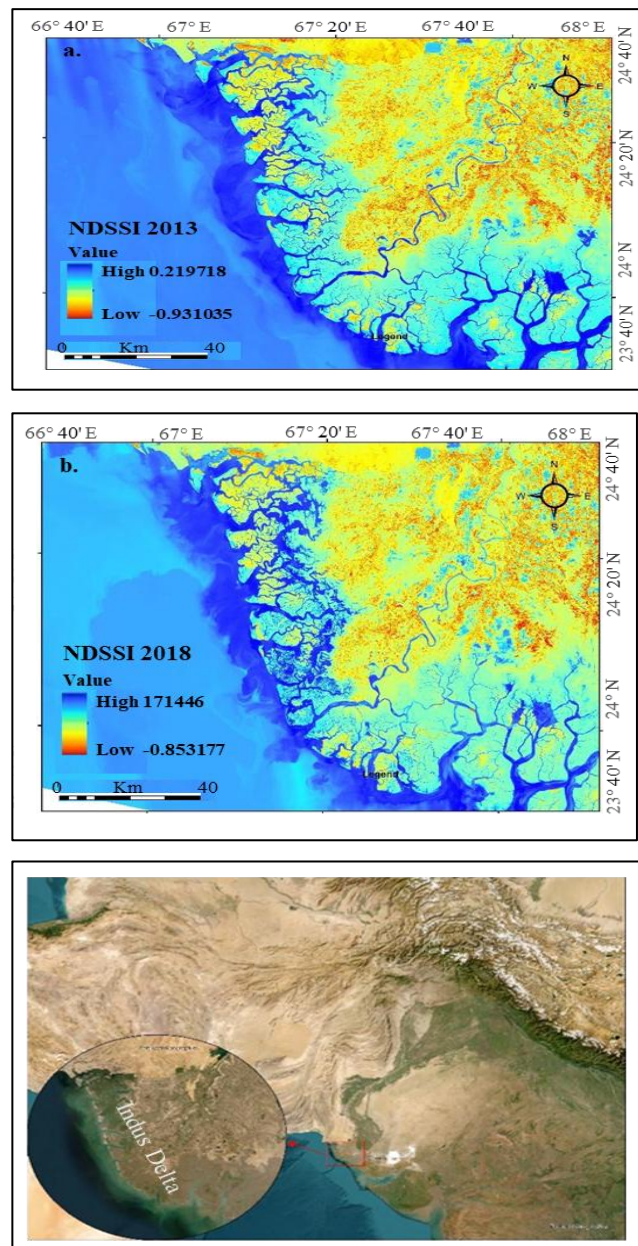


Fig 4. NDSSI map a. 2013, b. 2018 and c. 2023.

4.3 Vegetation Analysis

Normalized Difference Vegetation Index (NDVI) analysis for the years 2013, 2018, and 2023 is illustrated in Figure 5. The NDVI map depicts higher values up to 0.337 in the center and east regions of the delta, which is a clear indication of dense vegetation cover. The lower values are less than -0.274, represent sparse or degraded vegetation across the coast (Figure 5a). The total area of healthy vegetation in 2013 was approximately 4200.19 km² (Table 6). In 2018, the NDVI estimation is relatively higher than the results of 2013, where the assessed value is 0.471, a constructed rise in trend resulted due to government mangrove plantation and vegetation restoration initiatives (Figure 5b). A large part of the delta is still facing low values due to confined loss or degradation. By 2023 mix pattern has been revealed from NDVI analysis, a slight decline in value, i.e., 0.467, due to vegetation health, and a noticeable rise in NDVI observed along the coastline and inland regions Figure 5c. The overall decreasing trend and vegetation health highlight the ongoing degradation of the restored mangrove plantation project. Overall, the total “dense vegetation” area classified in 2013 was 4,200.19 km² has significantly reduced to 2,006.22 km² in 2018 and further reduced to 1,879.19 km² in 2023 (Table 6). The outcomes highlight the continued vulnerability of the deltaic ecosystem due to environmental and anthropogenic pressures.

Table 6

Vegetation Cover in distinct years

Year	Highest NDVI Value	Vegetation Condition	Dense Vegetation Area (km ²)
2013	0.337	Dense vegetation in the central & eastern delta with sparse/no vegetation near the coast	4,200.19
2018	0.471	Better vegetation due to mangrove plantation & some areas show degradation	2,006.22
2023	0.467		1,879.19

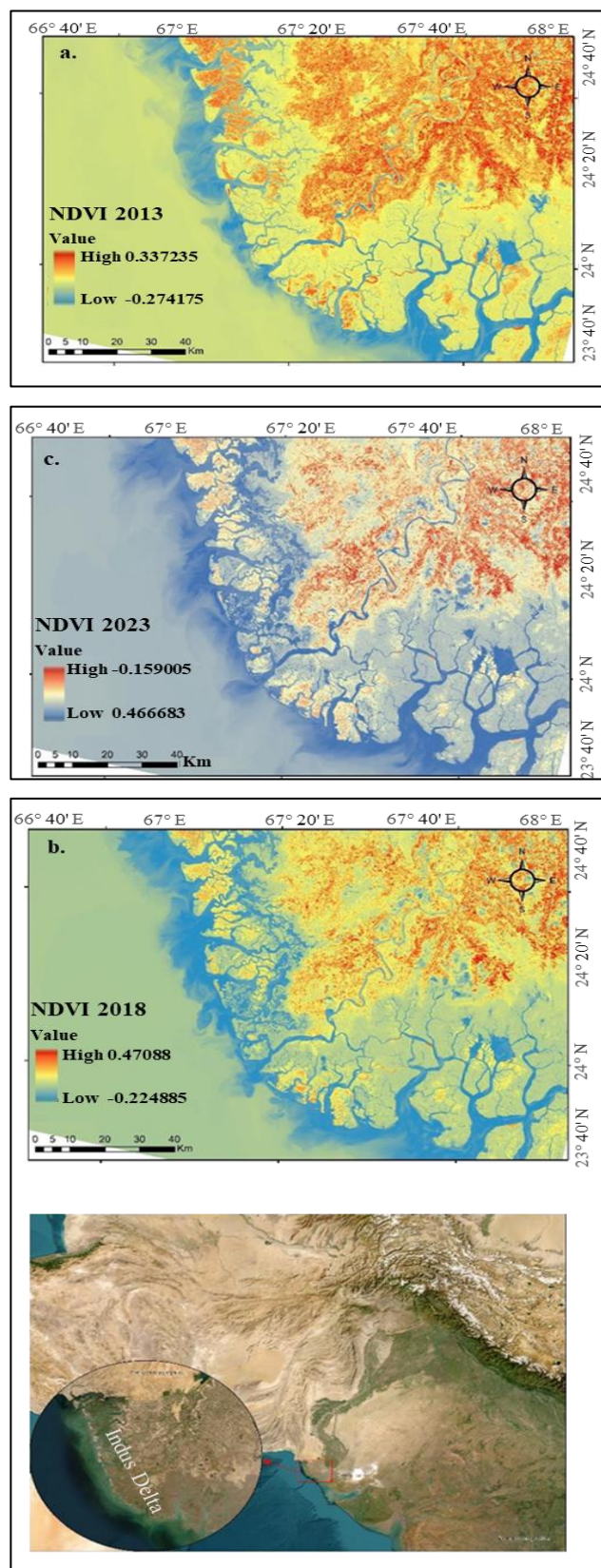


Fig. 5. Normalized Difference Vegetation Index (NDVI) map a. 2013, b. 2018, and c. 2023.

4.4 Coastal Erosion

Figure 7a shows the results of change detection, and the green color highlights “Erosion”, whereas the red color highlights the “Accretion” during all these years. Figure 7 (b & c) illustrates the results of the

Digital Shoreline Analysis System (DSAS) tool gives the following outputs: Net Shoreline Movement (NSM) and End Point Rate (EPR). The EPR map shows the rate of shoreline movement over a certain period, and it helps to understand the average annual rate of erosion or accretion. In the map, red to yellow zones indicate areas experiencing higher rates of erosion, with values ranging from -150.53 to -13.71 m/year, indicating significant shoreline retreat. Yellow to orange zones show lower rates of erosion or mild accretion, with values between -13.71 to 13.81 m/year, indicating relative stability. Green zones reflect areas of accretion with positive values ranging from 13.81 to 85.45 m/year, highlighting regions where sediments are depositing.

The NSM map represents the total span of shoreline displacement between the initial and recent shoreline positioning over the analysis period. In the map, the Red to blue zones have a value ranging from -796.02 to -119.32 meters, indicating extensive land loss. Yellow to orange zones show moderate shoreline movement, having values between -119.32 to 116.65 meters, suggesting relative stability with prominent movement. Green zones exhibit positive NSM values, indicating significant accretion with values ranging from 116.65 to 784.33 meters, highlighting regions of sediment deposition. Overall, the analysis reveals that the EPR rate is “-24.1283 m/year” and the NSM average NSM rate is “-226.42m” as shown in Table 8. The DSAS results highlighted several erosion hotspots along the Indus Delta; the DSAS tool's detailed summary is given in Table 8. On the other hand, areas with positive EPR and NSM values indicate regions where accretion is occurring, indicating regions of sediment deposition. Another technique of change detection is used to observe the change from 2013 to 2023. Environmental Indicator trends from 2013-2023 Indus Delta, Pakistan show the decreasing trend for dense vegetation, mangrove extent, NDVI, NDSSI, and suspended sediment load, whereas sea level is gradually rising in the study area (Figure 6).

Table 7

Decadal environmental indicators derived from Landsat 8/9 (2013 → 2023)

Indicator	2013	2018	2023	Net change 2013-23
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Dense-vegetation area (km ²)	4200.2	2006.2	1879.2	-55 %
Mangrove canopy extent (km ²)	1768	1521	1413	-20 %
NDVI (mean)	0.21	0.26	0.24	-0.02

NDSSI (mean)	0.131	0.112	0.098	-25 %
Suspended-sediment	38.4	31.2	28.8	-25 %
Altimetry-derived SLR (mm yr ⁻¹)	3.7	3.8	3.9	+0.2

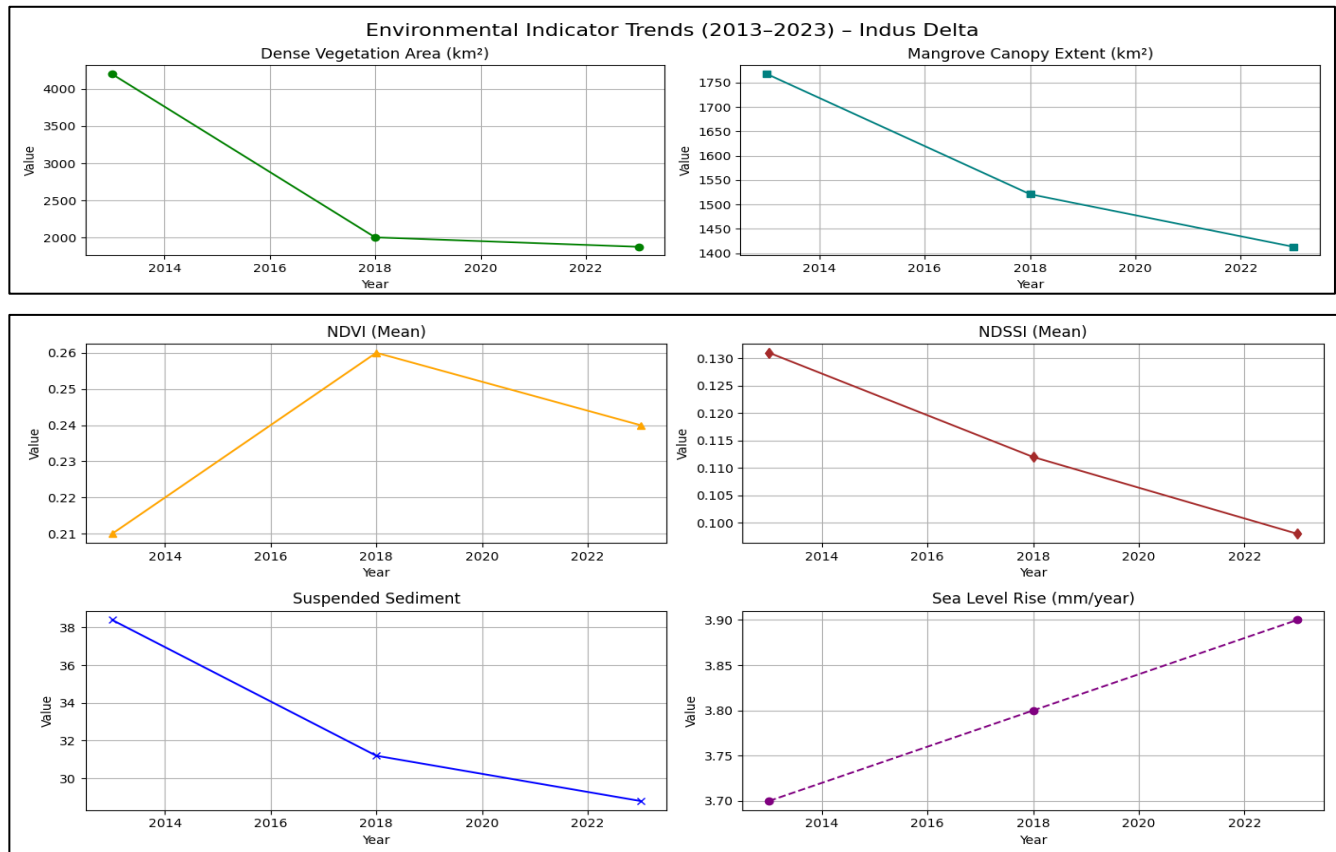


Fig. 6. Environmental Indicator trends from 2013-2023 Indus Delta, Pakistan

4.5 Shoreline Change with Statistical Confidence (DSAS Analysis)

The DSAS tool provided a detailed quantitative assessment of shoreline dynamics across ~2,800

Table 8

DSAS Shoreline Change Metrics with 95% Confidence Intervals (2013–2023)

Parameter	Mean	95% CI	Interpretation
Shoreline Change Envelope (SCE)	-278.25 m	± 19.45 m	Maximum landward shoreline movement (total range)
Net Shoreline Movement (NSM)	-226.42 m	± 16.87 m	Total displacement between the initial and final shorelines
End Point Rate (EPR)	-24.13 m/yr	± 2.12 m/yr	Annualized retreat rate using start & end positions
Linear Regression Rate (LRR)	-22.37 m/yr	± 2.41 m/yr	Trend-based estimate accounting for multiple dates

The confidence intervals (CI) indicate that shoreline erosion is not uniform. The mean EPR value was -24.13 m/year; some transects showed minor change within ±5 mm/yr, falling within natural

transects in the Indus Delta, capturing spatial and temporal variations from 2013 to 2023. The average shoreline metrics, along with 95% confidence intervals (CI), are summarized in Table 8.

variability or seasonal variations. Over 38% of transects exceed the upper CI margin, suggesting statistically significant, accelerated land loss.

The NSM in Figure 7b further illustrates these patterns. Positive values indicate deposition (accretion), though limited to protected sub-deltas.

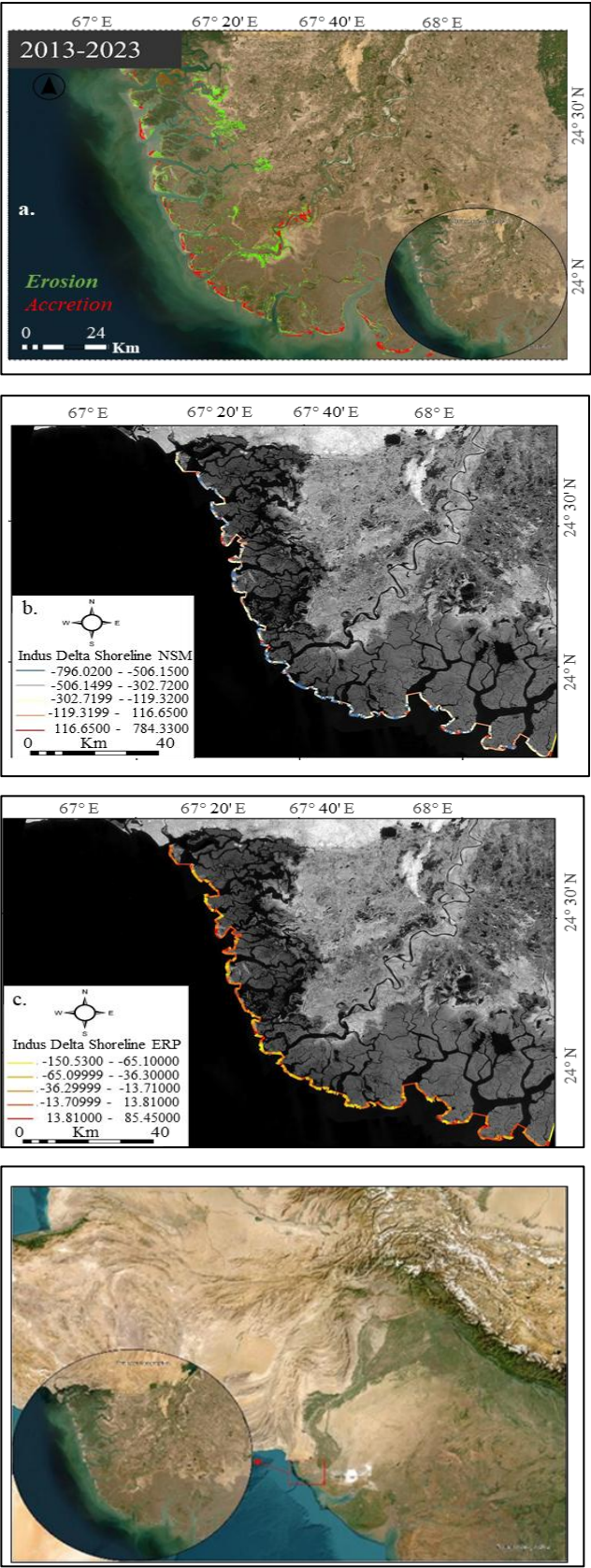


Fig 7 a. erosion and accretion map from 2013 to 2023, b. Indus delta shoreline NSM, c. Indus Delta Shoreline EPR.

The morpho-dynamics transition, accelerating SLR, Declining SSC, and pervasive erosion across the Indus Delta confirm the shift of the deltaic setting into

a wave-dominated system. The ecological consequences are also severe; dense and healthy vegetation has decreased by 55% removing a natural barrier against the storm surges. The analysis reveals that between 2013-2023, the Indus Delta experienced 84.4% erosion and 13.6% accretion.

5. Conclusion

This study is a comprehensive analysis of the various factors that contribute to the degradation of the Indus Delta. By utilizing distinct remote sensing tools and predictive models, this research has highlighted the significant change the delta has experienced from 2013 to 2023. The sea level rise (SLR) values aligned with IPCC projections of a global sea level rise. Sea level rise poses a significant threat to the delta, escalating the risks of coastal erosion and seawater intrusion on land. Furthermore, the estimation of the Normalized Difference Vegetation Index (NDVI) showed a declining trend in vegetation cover across the Indus Delta. This deterioration in vegetation is a critical indicator of the land degradation in the delta. The results are also supported by the NDSSI analysis for the same period highlighted the fluctuating trend of sediment dynamics within the delta, but generally, a decrease in SSC has been witnessed. Moreover, the Digital Shoreline Analysis System (DSAS) tool results indicated that coastal erosion and land loss are active in the area. Findings highlight that the combined influence of both climate change and modern human behaviors is depleting the Indus Delta. The continuous reduction in vegetation cover, interrupted sediment discharge, and a growing vulnerability to sea level rise all point to the depletion of the Indus delta. These results highlight the need for mitigation policies for environmental and anthropogenic factors to restore the Indus Delta. To safeguard the resilience of the Indus Delta, integrated conservation efforts, sustainable land use practices, and climate adaptation measures are essential to mitigate the adverse impacts and address UN SDGs (SDG 13-Climate Action and SDG 14-Life Below Water).

6. Recommendation

The integrated coastal management and monitoring to collect a continuous data set, necessary for future research on flood risk mapping, early warning, and vulnerability analysis. Pakistan Sea Level Data Portal, with the upgradation of existing tide gauges, combined with satellite detection and ground GPS data, provides an effective sea level trend and coastal risks. Assess uncertainties in sea-level rise forecasts using decision-making tools. Sea-level scenarios and decision-making frameworks should be integrated into development plans, infrastructure, and

coastal zone planning. Platforms should be developed to exchange knowledge and provide training to confirm resilience to sea level rise, coastal climate change, and marine conservation. The recommendation based on the current study is very critical to address the challenges triggered by sea level rise and enhance coastal resilience.

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IPCC	Intergovernmental Panel on Climate Change
SSC	Suspended Sediment Concentration
NFCS	National Framework for Climate Services
C3S	Copernicus Climate Change Service
CI	Confidence Interval
WMO	World Meteorological Organization
SLR (future)	Projected Sea Level Rise
σ	Standard Deviation
β_0, β_1	Regression Coefficients

Abbreviation	
GIS	Geographic Information System
RS	Remote Sensing
NDVI	Normalized Difference Vegetation Index
NDSSI	Normalized Difference Suspended Sediment Index
DSAS	Digital Shoreline Analysis System
SLR	Sea Level Rise
EPR	End Point Rate
NSM	Net Shoreline Movement
LRR	Linear Regression Rate
DEM	Digital Elevation Model
SCE	Shoreline Change Envelope
MAF	Million Acre Feet
OLI	Operational Land Imager
SRTM	Shuttle Radar Topography Mission
WRS	Worldwide Reference System